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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the May/June 2006 question paper

5014 ENVIRONMENTAL MANAGEMENT

5014/02 Paper 2, maximum raw mark 60

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1 **(a)** hot all year; many windy days; so evaporation rapid/speeded up/eq; **[1]**

(b) they carry out photosynthesis; which makes food/organic molecules for al… grow/provide food for consumers;

(c) energy lost /respiration loses energy; so only some food turned into consumer bodies/biomass; not enough to support more than two feeding levels; AVP; **[2]**

(d) **(i)** sustainable, salt water never runs out; process is solar powered; no significant pollution; AVP;
(ii) conservation, not many habitats for wading birds left; food chain can be maintained whilst evaporation goes on; AVP; **[max 3]**

2 **(a)** control building; build sewage treatment works; AVP; **[2]**

(b) **(i)** any three valid points, such as
leave no litter; do not kill fish; do not take coral away; report any damage; AVP;; **[3]**

(ii) 12; **[1]**

(iv) 4 shore and dive only, 4 boat shore and dive only; 4 boat shore dive and snorkel; AVP; **[max 3]**

(c) **(i)** nearly all boat access have higher % damage; highest shore damage 3.1% is only just higher than lowest boat damage 3.0%; **[2]**

(ii) more dives in total, explained; anchors do damage; boats hitting shallow corals; AVP;; **[3]**

(iii) 3-10; need for repetition to make findings more reliable/ref to means; **[2]**

(d) **(i)** cannot be exactly repeated; survey of only one post not enough; no distances specified; no undamaged corals recorded; **[max 3]**

(ii) correct orientation; length; all three posts; **[3]**

(iii) 135, 150; **[1]**

(iv) 15/150 x 100 = 10 (%); allow error from part **(iii)** **[1]**

(e) more damage/twice as much damage; around every post; AVP; **[2]**

(f) numbers of dives/divers at each survey; **[1]**

(g) some mention of all three aspects to score max 6 valid points;;;;;; **[6]**

3 **(a)** increased number of leaves; but reduced length; shorter plants less exposed to evaporation by wind; AVP; **[3]**

(b) **(i)** 8 m on scale; **[1]**

(ii) east-west line drawn; **[1]**

(iii) orientation;
axes labelled;
plot leaves;
plot length; **[4]**

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(iv) as distance increases from hedge number of leaves drops; and leng… drops;

(c) weigh fresh harvest from known sections of field/eq;

(d) two sensible densities/one may be lower than suggested;;; **[2]**

(e) soil exhaustion/minerals used up; loss of soil structure/more prone to erosion; disease in crops; loss of biodiversity in context; AVP;; **[3]**

Total 60